

Answers to Project Questions

1. Did we create a Delay feature?

No, in this project we did **not create a Delay feature**.

Instead, we created a different engineered feature called **Leadership_Gap**.

Leadership_Gap = PerformanceRating – JobSatisfaction

This feature was designed to represent situations where employee performance is high but job satisfaction is low, which may indicate potential **leadership or management issues** that could influence employee attrition.

2. Was the alternative feature (Leadership_Gap) applied in all models?

Yes.

The **Leadership_Gap feature** was included in the feature set used to train all machine learning models:

- Random Forest
- Logistic Regression
- Neural Network

This means that every model used the same input features when making predictions, ensuring a fair comparison between models.

3. Why didn't we calculate Feature Importance for Neural Network and Logistic Regression?

Feature importance was only calculated for the **Random Forest model**.

This is because:

- **Random Forest** is a tree-based model that naturally provides feature importance scores.
- These scores show how much each feature contributes to the prediction.

However:

- **Neural Networks** are considered **black-box models**, meaning it is difficult to directly interpret which features are most important.
- **Logistic Regression** can provide coefficients, but they are not always as intuitive or stable as feature importance in tree-based models.

Therefore, Random Forest was used to analyze which variables most strongly influence employee attrition.

4. Which model achieved the highest accuracy?

The **Random Forest model achieved the highest accuracy** among the tested models.

Reasons include:

- It combines multiple decision trees.
- It can capture **complex and non-linear relationships** between variables.
- It is robust to noise and variations in the dataset.

As a result, Random Forest performed best in predicting employee attrition.

5. Is it normal for Confusion Matrices to be different between models?

Yes, it is completely normal.

Each machine learning model uses a **different learning algorithm**, which leads to slightly different predictions.

Because predictions differ, the **Confusion Matrix values will also differ**.

However, the **Actual values should be the same** because all models are evaluated on the **same test dataset**.

We can verify this by summing the rows in the confusion matrices.

Example:

Random Forest

```
[[35, 0],
 [0, 25]]
```

Total Actual Values:

Actual Class 0 $\rightarrow$ 35

Actual Class 1 $\rightarrow$ 25

Neural Network

[[34, 1],

[1, 24]]

Total Actual Values:

Actual Class 0 $\rightarrow$ $34 + 1 = \mathbf{35}$

Actual Class 1 $\rightarrow$ $1 + 24 = \mathbf{25}$

This confirms that the **actual data distribution remains the same**, while only the predictions differ between models.